
EviSAM3: Evidence-Driven SAM3 for Referring Remote Sensing Image Segmentation

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Abstract

Referring Remote Sensing Image Segmentation (RRSIS) aims to segment target objects from remote sensing images based on natural language descriptions, requiring precise vision–language alignment under significant ambiguity. Despite the strong generalization ability of Segment Anything Model 3 (SAM3), the large spatial extent and varying scales of targets in remote sensing images intensify the ambiguity of referring expressions. This poses a major challenge for fine-grained segmentation. To address the challenge, we propose **EviSAM3**, an evidence-driven parameter-efficient adaptation framework that mimics human decision-making by dynamically accumulating partial evidence. Specifically, EviSAM3 progressively integrates cross-modal evidence through an evidence memory bank, where each incoming piece is aggregated via momentum-based updates. As evidence accumulates, the model performs iterative evidence rectification to resolve ambiguity under incomplete observations. Meanwhile, the contributions of different evidence are adaptively modulated based on their diagnostic relevance, allowing more informative evidence to dominate the reasoning process and guide the final prediction. Extensive experiments demonstrate that EviSAM3 consistently outperforms state-of-the-art methods, particularly in challenging scenarios with high ambiguity and significant scale variation.

1 Introduction

Referring Remote Sensing Image Segmentation (RRSIS) is the task of segmenting target regions in remote sensing images based on natural language descriptions Rong et al. [2026], Zhang et al. [2026], Yuan et al. [2024]. It can be viewed as a specialized subtask of Referring Image Segmentation (RIS), focusing on the unique properties of remote sensing imagery Mao et al. [2025], Liu et al. [2025c], Hu et al. [2023], Liu and Li [2025]. This task holds significant practical potential and can be applied to various real-world scenarios, including disaster assessment Mi et al. [2025], urban analysis Wellmann et al. [2020], and environmental monitoring Dai et al. [2023].

Recently, Segment Anything Model 3 (SAM3) Kirillov et al. [2023], Ravi et al. [2025], Carion et al. [2026] has demonstrated strong generalization ability in generic image segmentation, benefiting

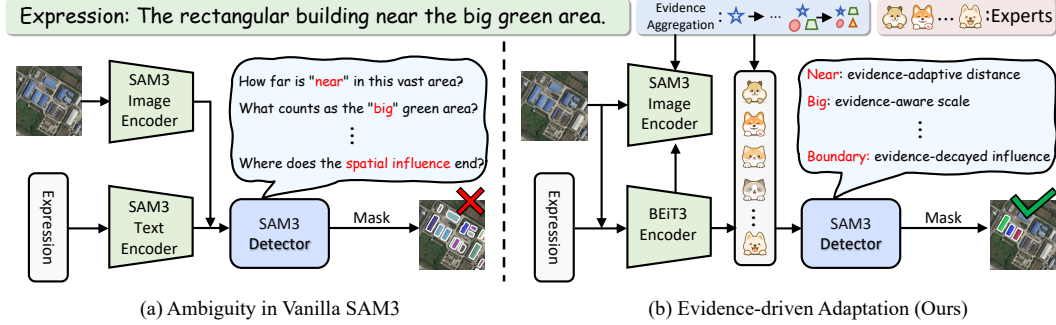


Figure 1: Comparison between (a) the vanilla SAM3-based baseline and (b) our proposed EviSAM3.

from large-scale pre-training and its promptable design. Subsequent studies have further explored its adaptability across diverse domains, including medical image segmentation Liu et al. [2025a], camouflage object segmentation Chen et al. [2025b], and text-guided annotation systems, highlighting its strong transferability and extensibility Seok et al. [2026]. Despite these advances, directly applying SAM3 to RRSIS remains challenging. Remote sensing imagery involves large spatial extents and extreme scale variations, which amplify the ambiguity of referring expressions and hinder precise vision–language alignment.

Despite its strong generalization ability, SAM3 fundamentally relies on concept-level prompts, which are typically represented as coarse noun phrases Carion et al. [2026], Sapkota et al. [2025]. This design prevents it from modeling structured referring expressions involving spatial relations, relative attributes, and contextual reasoning. As illustrated in Fig. 1a, spatial descriptions such as "near" and "big" lack explicit and consistent definitions in large-scale remote sensing imagery. As illustrated in Fig. 1(b), spatial terms such as "near" and "big" are inherently ambiguous, lacking consistent definitions across scenes and scales. Such ambiguity leads to inconsistent and unreliable prompt interpretation, causing SAM3 to produce inaccurate segmentation results. More critically, SAM3 performs category-level matching that identifies all instances of a category, rather than grounding a specific instance described by relational or contextual cues. This paradigm mismatch makes it fundamentally unsuitable for RRSIS, where the target is often defined by ambiguous and context-dependent descriptions rather than explicit semantic categories.

To overcome this limitation, we investigate how humans resolve ambiguous referring expressions in complex visual scenes and identify target objects under uncertainty. Drawing on insights from Evidence Accumulation Models (EAMs) in cognitive psychology Lee and Cummins [2004], Davidson and El Hady [2019], Evans et al. [2020], we view this process not as instantaneous text–image matching, but as a progressive decision-making procedure where evidence is accumulated over time. From a cognitive perspective, humans do not directly map vague descriptions to precise regions, but instead rely on partial cues that are gradually integrated to refine their understanding. Concretely, when humans handle ambiguous referring expressions in remote sensing imagery, the process can be understood as a gradual integration of evidence rather than a direct mapping. At the beginning, they rely on coarse semantic cues such as "building" or "green area" to roughly narrow down the search space, but these cues alone are often insufficient due to category redundancy and large intra-class variation. As the process continues, more specific information, including spatial relations like "near", relative attributes such as "big", and surrounding context, is progressively taken into account. Not all cues play the same role, and people tend to rely more on those that are more informative, while naturally downplaying less relevant information.

Based on the above motivations, we propose Evidence-Driven SAM3 (**EviSAM3**), a framework that enables SAM3 to interpret ambiguous referring expressions in remote sensing imagery through progressive evidence reasoning. As illustrated in Fig. 1b, instead of relying on single-step prompt interpretation, EviSAM3 models grounding as the gradual integration of multiple cues, progressively refining target localization. We design an Evidence Integration Module (**EIM**) to model the integration of visual–linguistic evidence. EIM maintains a memory bank of intermediate evidence and iteratively infers new evidence through cross-modal interactions. The inferred evidence is then combined with historical evidence via a progressive update mechanism, enabling more reliable interpretation of ambiguous spatial descriptions and contextual cues. Inspired by Mixture-of-Experts architectures

in large language models, where multiple experts specialize in different knowledge patterns Dai et al. [2024], Cai et al. [2025], Chen et al. [2022], we further design an Evidence-Driven Mixture-of-Experts (**ED-MoE**) Prompting strategy to convert accumulated evidence into segmentation guidance. ED-MoE dynamically selects specialized experts for different visual patterns, such as structural details, cluttered scenes, and spatial configurations, and produces refined prompts to guide the SAM3 mask decoder for accurate segmentation. The contributions are summarized as follows:

- We analyze the limitations of the vanilla SAM3 when applied to RRSIS and reveal that its concept-level prompting paradigm oversimplifies referring segmentation as a category matching problem, which fails to handle ambiguous expressions involving spatial relations, relative attributes, and contextual cues.
- We propose **EviSAM3**, an evidence-driven framework that reformulates referring segmentation as a progressive evidence reasoning process, enabling SAM3 to gradually refine target localization under ambiguous descriptions.
- We design an Evidence Integration Module (**EIM**) to progressively integrate visual–linguistic evidence through a memory-based update mechanism, and design an Evidence-Driven Mixture-of-Experts (**ED-MoE**) Prompting strategy that dynamically routes evidence-aware features to specialized experts for generating refined segmentation prompts.
- Extensive experiments on RRSIS benchmarks demonstrate that EviSAM3 effectively resolves referential ambiguity and significantly outperforms state-of-the-art methods.

2 Related Work

Referring Remote Sensing Image Segmentation. RRSIS is an extension of Referring Image Segmentation (RIS) Mao et al. [2025], Xia et al. [2024], Ding et al. [2022], Wu et al. [2024], Chng et al. [2024], Liu et al. [2025b]. Early work Yuan et al. [2024], Peng et al. [2025], Li et al. [2025c] established the RefSegRS dataset and introduced the language-guided cross-scale enhancement (LGCE) module to improve the detection of small and scattered objects. To address challenges such as large scale variations and complex orientations in aerial imagery, subsequent methods have explored more effective multi-scale and cross-modal interactions. For instance, RMSIN Liu et al. [2024b] incorporates adaptive rotated convolutions for scale-aware feature interaction, while SBANet Li et al. [2025b] models scale-wise bidirectional alignment to better capture global and local information. Later approaches, including CroBIM Dong et al. [2024] and DANet Pan et al. [2024], further enhance cross-modal alignment through iterative interaction and explicit discrepancy reduction. More recently, the emergence of foundation models has driven new frameworks such as RSRefSeg Chen et al. [2025a], which leverages CLIP Radford et al. [2021] and SAM Kirillov et al. [2023] for improved generalization, and RS2-SAM2 Rong et al. [2026], which adapts SAM2 Ravi et al. [2025] to remote sensing via hierarchical fusion strategies. While existing methods achieve promising results, they largely rely on one-step vision–language alignment, limiting their ability to handle ambiguity and complex spatial reasoning in remote sensing scenarios.

Segment Anything Model. The Segment Anything Model (SAM) Kirillov et al. [2023] introduced a promptable segmentation paradigm with strong zero-shot transfer on the billion-mask SA-1B dataset. SAM 2 Ravi et al. [2025], Wang et al. [2026] extended it to video via a streaming memory, while SAM 3 Meta AI [2025] further replaces geometric prompts with concept prompts, segmenting all instances matching a noun phrase or exemplar. Since SAM variants are pretrained on natural images, several works adapt them to remote sensing through task-specific prompting Chen et al. [2024], domain adapters Zhang et al. [2025], Li et al. [2024], parameter-efficient tuning Zhong et al. [2024], Houlsby et al. [2019], Han et al. [2024], or SAM 3-based open-vocabulary aerial segmentation Li et al. [2025a]. HQ-SAM improves fine-grained mask quality, while PerSAM enables one-shot personalization for target-specific segmentation Ke et al. [2023], Zhang et al. [2023]. However, these methods still treat textual prompts as deterministic signals, leaving the cross-modal ambiguity of aerial referring expressions unmodeled. This motivates a closer look at how uncertainty can be incorporated into SAM-based referring segmentation.

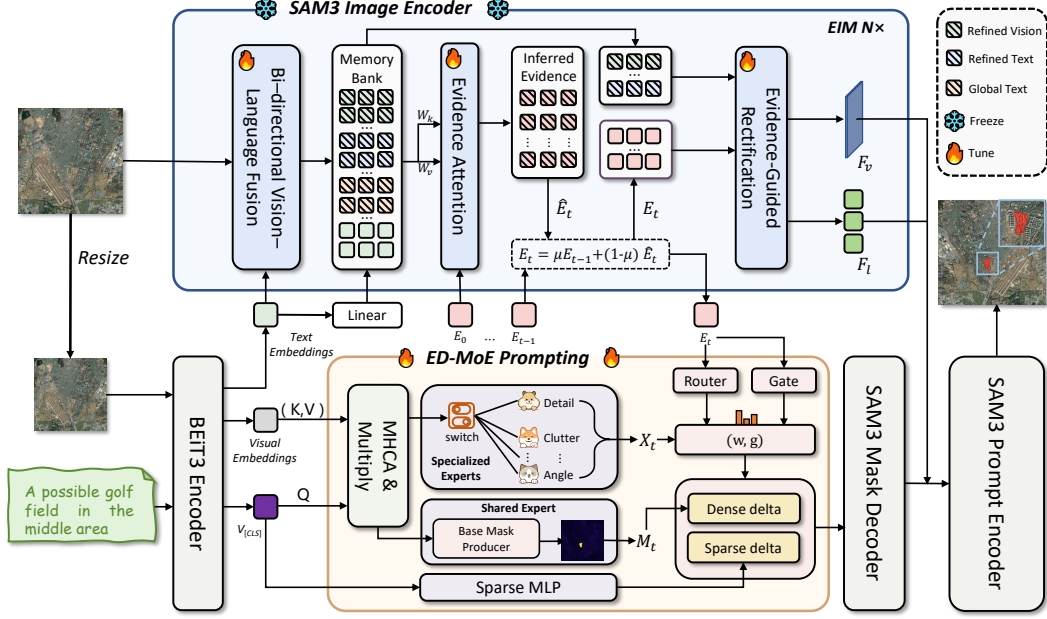


Figure 2: Overview of the proposed EviSAM3 framework. The Evidence Integration Module (EIM) progressively fuses visual and textual cues across backbone layers, accumulates intermediate evidence, and rectifies feature representations to reduce ambiguity in referring expressions. The Evidence-Driven Mixture-of-Experts (ED-MoE) module then converts the accumulated evidence into specialized prompt refinements, enabling the decoder to emphasize boundary details, elongated structures, and cluttered scene patterns for accurate final segmentation.

3 Method

3.1 Overview

Given a remote sensing image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ and a referring expression $\mathbf{E} = \{e_l\}_{l=1}^L$, the task is to predict a binary mask $\mathbf{M} \in \{0, 1\}^{H \times W}$ for the referred target. As shown in Fig. 2, EviSAM3 follows a two-stage reasoning pipeline: (1) *evidence integration* in a mostly frozen SAM3 backbone Carion et al. [2026], and (2) *evidence-driven prompt generation* for mask decoding.

We first extract compact multimodal priors from a pretrained vision-language encoder Zhang et al. [2024], Rong et al. [2026], using a low-resolution image (224×224) and tokenized text. This encoder outputs a multimodal class token $\mathbf{v}_{\text{cls}}$, visual embeddings $\mathbf{V}$, and text embeddings $\mathbf{T}$. The class token is fused with visual tokens by cross-attention followed by element-wise interaction: $\mathbf{v}'_{\text{cls}} = \mathbf{v}_{\text{cls}} \odot \text{MHCA}(\mathbf{v}_{\text{cls}}, \mathbf{V}, \mathbf{V})$, which provides the base signal for downstream prompt construction. Meanwhile, $\mathbf{T}$ is projected to $\mathbf{T}' \in \mathbb{R}^{N_t \times d}$ ($d=256$) and fed into adapter blocks for progressive cross-modal evidence reasoning. The final dense and sparse prompts are then decoded by the SAM3 mask decoder Carion et al. [2026].

3.2 Evidence Integration Module

While the multimodal prior extractor provides a joint vision-language representation, the SAM3 backbone operates on high-resolution images (1008×1008) without direct access to textual inputs. To enable progressive cross-modal reasoning within the mostly frozen backbone, we inject an *Evidence Rectified Adapter* (ERA) into each of the $N_b = 32$ ViT blocks.

ERA Formulation. For each block i , let $\mathbf{F}_i \in \mathbb{R}^{H \times W \times d}$ and $\mathbf{T}_i \in \mathbb{R}^{d_t}$ denote the visual and text features, respectively. ERA first infers a spatial evidence map via cross-modal interaction: $\mathbf{E}_i = A_e(\text{xattn}(\mathbf{F}_i, \mathbf{T}_i)) \in \mathbb{R}^{H \times W}$, where $\text{xattn}(\cdot)$ denotes cross-attention and $A_e(\cdot)$ projects features to a spatial evidence map. Based on the inferred evidence, ERA performs gated residual

rectification:

$$\mathbf{F}'_i = \mathbf{F}_i + \sigma(g_{v,i})\lambda_{v,i}(W_v\mathbf{F}_i) \odot \mathbf{E}_i, \quad \mathbf{T}'_i = \mathbf{T}_i + \sigma(g_{t,i})\lambda_{t,i}U(\mathbf{T}_i, \mathbf{E}_i), \quad (1)$$

where $g_{v,i}, g_{t,i}$ are learnable gates, $\lambda_{v,i}, \lambda_{t,i}$ are LayerScale parameters, and $\sigma(\cdot)$ is the sigmoid function. This formulation characterizes ERA as an evidence-guided residual modulation mechanism. In practice, we instantiate this formulation with three components: bidirectional vision–language fusion, progressive evidence accumulation, and evidence-based feature rectification.

Bidirectional Vision–Language Fusion. Both modalities are first projected into a shared hidden space ($D_h = C/4$). We then perform bidirectional cross-attention:

$$\hat{\mathbf{F}}_i = \text{MHCA}(\mathbf{F}'_i, \mathbf{T}'_{h,i}) + \mathbf{F}'_i, \quad \hat{\mathbf{T}}_{h,i} = \text{MHCA}(\mathbf{T}'_{h,i}, \mathbf{F}'_i) + \mathbf{T}'_{h,i}. \quad (2)$$

Progressive Evidence Accumulation. We maintain evidence in a latent token space and progressively update it across layers. At block i , we compute posterior evidence $\tilde{\mathbf{e}}_i$ using learnable queries $\mathbf{q} \in \mathbb{R}^{E \times d}$ over a memory bank:

$$\mathbf{m}_i = [\mathbf{T}_i; \phi_t(\hat{\mathbf{T}}_{h,i}); \phi_v(\hat{\mathbf{F}}_i)], \quad \tilde{\mathbf{e}}_i = \text{MHCA}(\mathbf{q}, \mathbf{m}_i, \mathbf{m}_i), \quad (3)$$

where ϕ_t and ϕ_v are linear projections. Evidence is accumulated via exponential moving average:

$$\mathbf{e}_i = \mu\mathbf{e}_{i-1} + (1 - \mu)\tilde{\mathbf{e}}_i, \quad \mu = 0.6, \quad (4)$$

allowing gradual integration of low-level and high-level cues.

Evidence Rectification. The accumulated evidence is projected back to the hidden space and applied to refine features:

$$\hat{\mathbf{F}}_i \leftarrow \hat{\mathbf{F}}_i + \sigma(g_v) \cdot (\text{LN}(\text{MHCA}(\hat{\mathbf{F}}_i, \psi(\mathbf{e}_i), \psi(\mathbf{e}_i))) \odot \lambda_v), \quad (5)$$

with an analogous update for the text branch. The gates are conservatively initialized (e.g., $g_v = -6$, $\lambda_v = 10^{-3}$) to preserve pretrained representations while enabling gradual evidence-driven refinement. Finally, an MLP adapter is applied via residual connection ($\alpha = 0.5$).

3.3 Evidence-Driven Mixture-of-Experts Prompting and Decoding

After all N_b blocks, the accumulated evidence $\bar{\mathbf{e}}$ is aggregated from the available backbone output layers (in our default setting, this is primarily the last output layer). The base mask logits are computed by element-wise multiplying the fused class token $\mathbf{v}'_{\text{cls}}$ with the reshaped visual embeddings $\mathbf{V}_{2D}$, followed by linear projection: $\mathbf{L}_{\text{base}} = \text{Linear}(\mathbf{V}_{2D} \odot \mathbf{v}'_{\text{cls}})$.

Expert Routing and Specialization. ED-MoE routes the evidence to three specialized experts: (1) a **Detail Expert** using 3×3 depthwise convolution for fine-grained boundaries; (2) an **Elongated Expert** using Adaptive Rotated Convolution (ARC) Liu et al. [2024b] for oriented structures (e.g., roads, rivers); and (3) a **Clutter Expert** using GroupNorm + 1×1 convolution for cluttered backgrounds. The routing weights are:

$$\mathbf{r} = [\bar{\mathbf{e}}_{\text{pool}}; \beta \cdot \mathbf{t}_{\text{cls}}], \quad \mathbf{w} = \text{softmax}(\text{MLP}(\mathbf{r})/\tau), \quad (6)$$

where $\bar{\mathbf{e}}_{\text{pool}}$ is the token-averaged evidence, $\mathbf{t}_{\text{cls}}$ is the projected text class embedding, β is a learnable coefficient (initialized to 0), and $\tau=1.5$ is the temperature. Each expert produces dense logit deltas $\Delta\mathbf{L}_k$ and token deltas $\Delta\mathbf{s}_k$, which are weighted-summed and gated to form the final outputs:

$$\mathbf{L} = \mathbf{L}_{\text{base}} + g \cdot \gamma \sum_k w_k \Delta\mathbf{L}_k, \quad \mathbf{S} = [\mathbf{s}_{\text{cls}}; g \cdot \gamma' \sum_k w_k \Delta\mathbf{s}_k], \quad (7)$$

where $g = \sigma(\text{Linear}(\mathbf{r}))$ is a sigmoid gate (initialized near zero), and γ, γ' are LayerScale parameters (10^{-3}). A balance loss $\mathcal{L}_{\text{bal}} = K \sum_k \bar{w}_k^2$ Dai et al. [2024] encourages uniform expert utilization.

The final logits $\mathbf{L}$ are converted into dense mask prompts via sigmoid and a lightweight projection head, then combined with sparse prompt tokens $\mathbf{S}$ and decoded by the SAM3 mask decoder Carion et al. [2026]. The decoder performs bidirectional interaction between prompts and multi-scale features (three resolutions when high-resolution features are enabled). Training uses

$$\mathcal{L} = \lambda_{\text{ce}}\mathcal{L}_{\text{ce}} + \lambda_{\text{dice}}\mathcal{L}_{\text{dice}} + \lambda_{\text{bal}}\mathcal{L}_{\text{bal}}, \quad (8)$$

with $\lambda_{\text{ce}}=0.9$, $\lambda_{\text{dice}}=0.1$, and $\lambda_{\text{bal}}=0.01$ Milletari et al. [2016], Dai et al. [2024]. Most SAM3 backbone parameters are frozen except the prompt generator; the prior extractor, adapters, and ED-MoE are optimized with differential learning rates, while the remaining trainable parameters use the base learning rate.

Table 1: Comparison with state-of-the-art methods on the RRSIS-D dataset. The best and second-best results are highlighted in **bold** and underlined, respectively.

Method	Pr@0.5		Pr@0.6		Pr@0.7		Pr@0.8		Pr@0.9		oIoU		mIoU	
	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
CMPC Huang et al. [2020]	57.93	55.83	48.85	47.40	38.50	36.94	25.28	25.45	9.31	9.19	70.15	69.22	50.41	49.24
BRINet Hu et al. [2020]	58.79	56.90	49.54	48.77	39.65	39.12	28.21	27.03	9.19	8.73	70.73	69.88	51.14	49.65
CMPC+ Liu et al. [2021]	59.19	57.65	49.36	47.51	38.67	36.97	25.91	24.33	8.16	7.78	70.14	68.64	51.41	50.24
LGCE Yuan et al. [2023]	68.10	67.65	60.52	61.53	52.24	51.45	42.24	39.62	23.85	23.33	76.68	76.34	60.16	59.37
RIS-DMMI Hu et al. [2023]	70.40	68.74	63.05	60.96	54.14	50.33	41.95	38.38	23.85	21.63	77.01	76.20	60.72	60.12
LAVT Yang et al. [2022]	69.54	69.52	63.51	63.63	53.16	53.29	43.97	41.60	24.25	24.94	77.59	77.19	61.46	61.04
EVF-SAM Zhang et al. [2024]	73.51	72.16	67.87	66.50	58.33	56.59	46.15	43.92	25.92	25.48	76.32	76.77	64.03	62.75
FIANet Lei et al. [2025]	-	74.46	-	66.96	-	56.31	-	42.83	-	24.13	-	76.91	-	64.01
RMSIN Liu et al. [2024a]	74.66	74.26	68.22	67.25	57.41	55.93	45.29	42.55	24.43	24.53	78.27	77.79	65.10	64.20
RS2-SAM2 Rong et al. [2025]	<u>79.25</u>	<u>77.56</u>	<u>77.56</u>	<u>72.34</u>	<u>63.85</u>	<u>63.85</u>	<u>50.57</u>	<u>47.92</u>	<u>30.40</u>	<u>29.73</u>	<u>80.16</u>	<u>78.99</u>	<u>68.81</u>	<u>66.72</u>
SAM3-Adapter Chen et al. [2025b]	72.31	73.65	65.19	66.69	55.50	56.08	43.06	40.80	23.28	22.59	74.47	74.44	63.71	63.39
SAM3-LoRA Hu et al. [2022]	67.26	67.14	60.95	60.26	51.09	50.37	40.54	38.99	22.13	22.16	72.63	73.03	59.99	59.81
EviSAM3 (Ours)	83.45	82.81	77.82	77.65	66.95	66.37	53.33	50.56	32.36	30.82	81.38	80.27	71.25	70.65

4 Experiment

We conduct a comprehensive evaluation of EviSAM3 to address the following research questions: **RQ1)** how does EviSAM3 perform compared to state-of-the-art methods? **RQ2)** what is the contribution of each component? **RQ3)** how effective is the evidence accumulation mechanism under ambiguous expressions? **RQ4)** what are its limitations in challenging scenarios?

4.1 Experimental Settings

Datasets. We evaluate our method on two representative RRSIS benchmarks, RefSegRS Yuan et al. [2024] and RRSIS-D Liu et al. [2024b], capturing both semantic and spatial challenges in remote sensing imagery, as well as the inherent ambiguity in referring expressions. RefSegRS consists of 512×512 samples with diverse scenes and compositional expressions, emphasizing fine-grained alignment under appearance similarity and scale variation. RRSIS-D, with higher-resolution images (800×800), introduces dense instances, arbitrary orientations, and elongated structures, along with complex background clutter, enabling a comprehensive evaluation of model robustness.

Implementation Details. All experiments are conducted on 4 NVIDIA H200 GPUs with a batch size of 4 per GPU, 512GB system memory, and the Ubuntu operating system. We train for 60 epochs on both RefSegRS and RRSIS-D. The AdamW optimizer Loshchilov and Hutter [2017] is employed with a total batch size of 16. The base learning rates are set to 5×10^{-5} for RefSegRS and 1×10^{-5} for RRSIS-D. In the final 10 epochs, all learning rates are decayed to 0.1 of their initial values.

Metrics. Following the evaluation protocol of RS2-SAM2 Rong et al. [2025], we measure performance using multiple criteria, including precision under different IoU thresholds (Pr@0.5–Pr@0.9), mean Intersection-over-Union (mIoU), and overall Intersection-over-Union (oIoU). All metrics are reported on the validation and test splits of both RefSegRS and RRSIS-D.

4.2 Comparison with State-of-the-Art Methods (RQ1)

RRSIS-D dataset. As shown in Table 1, EviSAM3 consistently outperforms all prior methods. On the test set, it surpasses the strongest baseline RS2-SAM2 Rong et al. [2025] by +5.25 at Pr@0.5 and maintains gains at stricter thresholds (e.g., +1.09 at Pr@0.9), indicating more precise localization under high-overlap criteria. In terms of region-level metrics, EviSAM3 achieves 80.27 oIoU and 70.65 mIoU, improving over RS2-SAM2 Rong et al. [2025] by +1.28 and +3.93, respectively, with the larger gain in mIoU reflecting better overall mask quality. Compared with earlier RIS methods such as LAVT Yang et al. [2022] and RMSIN Liu et al. [2024a], the margin is substantial, highlighting the difficulty of handling complex spatial layouts. Furthermore, EviSAM3 also consistently outperforms SAM-based adaptations, including SAM3-Adapter Chen et al. [2025b] and SAM3-LoRA Hu et al. [2022], suggesting that simple parameter-efficient tuning is insufficient for resolving ambiguity in RRSIS.

RefSegRS dataset. As shown in Table 2, EviSAM3 significantly outperforms all prior methods. Compared with RS2-SAM2 Rong et al. [2025], it improves Pr@0.5 by +5.62 and achieves larger gains at higher thresholds, reaching +12.93 at Pr@0.8 and +24.76 at Pr@0.9, indicating markedly

Table 2: Comparison with state-of-the-art methods on the RefSegRS dataset. The best and second-best results are highlighted in **bold** and underlined, respectively.

Method	Pr@0.5		Pr@0.6		Pr@0.7		Pr@0.8		Pr@0.9		oIoU		mIoU	
	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
BRINet Hu et al. [2020]	36.86	20.72	35.53	14.26	19.93	9.87	10.66	2.98	2.84	1.14	61.59	58.22	38.73	31.51
EVF-SAM Zhang et al. [2024]	57.77	35.17	37.59	22.34	16.24	9.36	4.87	2.86	1.86	0.39	59.61	55.51	46.98	36.64
CMPC Huang et al. [2020]	46.09	32.36	26.45	14.14	12.76	6.55	7.42	1.76	1.39	0.22	63.55	55.39	42.08	40.63
CMPC+ Liu et al. [2021]	56.84	49.19	37.59	28.31	20.42	15.31	10.67	8.12	2.78	2.55	70.62	66.53	47.13	43.65
CARIS Liu et al. [2023]	68.45	45.40	47.10	27.19	25.52	15.08	14.62	8.87	3.71	1.98	75.79	69.74	54.30	42.66
LAVT Yang et al. [2022]	80.97	51.84	58.70	30.27	31.09	17.34	15.55	9.52	4.64	2.09	78.50	71.86	61.53	47.40
RIS-DMMI Hu et al. [2023]	86.17	63.89	74.71	44.30	38.05	19.81	18.10	6.49	3.25	1.00	74.02	68.58	65.72	52.15
LGCE Yuan et al. [2023]	90.72	73.75	86.31	61.14	71.93	39.46	32.95	16.02	10.21	5.45	83.56	76.81	72.51	59.96
RMSIN Liu et al. [2024a]	93.97	79.20	89.33	65.99	74.25	42.98	29.70	16.51	7.89	3.25	82.41	75.72	73.84	62.58
FIANet Lei et al. [2025]	-	84.09	-	77.05	-	61.86	-	33.41	-	7.10	-	78.32	-	68.67
RS2-SAM2 Rong et al. [2025]	<u>95.36</u>	<u>84.31</u>	<u>94.90</u>	<u>79.42</u>	<u>92.58</u>	<u>70.89</u>	<u>83.76</u>	<u>55.70</u>	<u>36.66</u>	<u>21.19</u>	<u>88.03</u>	<u>80.87</u>	<u>80.87</u>	<u>73.90</u>
SAM3-Adapter Chen et al. [2025b]	81.21	66.70	74.48	56.74	63.57	43.20	42.23	20.25	9.05	4.40	72.52	66.62	69.09	58.54
SAM3-LoRA Hu et al. [2022]	86.77	67.58	78.89	56.52	63.11	41.50	37.82	16.46	8.12	4.07	79.28	73.89	70.35	58.11
EviSAM3 (Ours)	98.14	89.93	96.29	85.25	94.90	79.58	93.27	68.63	83.53	45.95	92.44	86.51	91.32	80.56

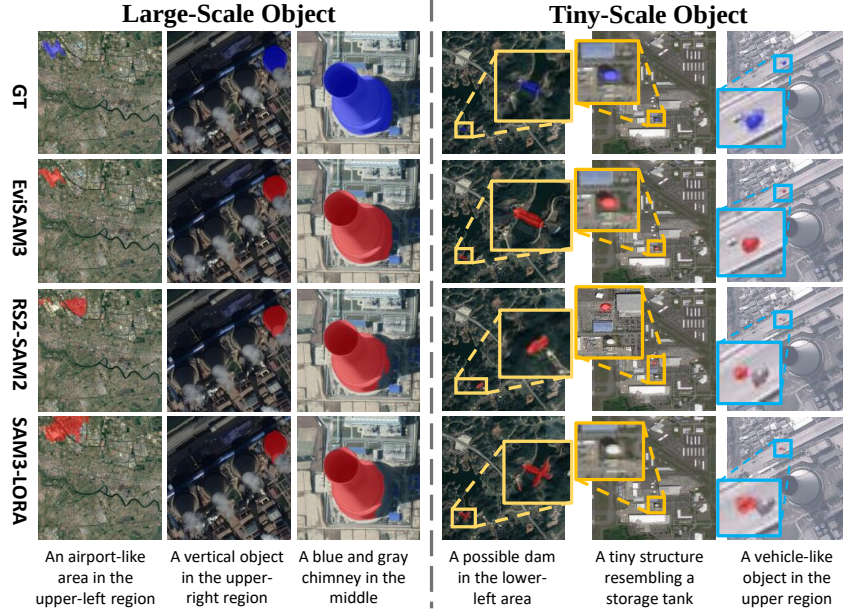


Figure 3: Qualitative comparison results on multi-scale targets under ambiguous referring expressions.

better high-precision segmentation. In terms of region-level metrics, EviSAM3 surpasses RS2-SAM2 Rong et al. [2025] by +5.64 oIoU and +6.66 mIoU, reflecting more consistent mask quality. It also clearly outperforms earlier methods such as LAVT Yang et al. [2022], LGCE Yuan et al. [2023], and RMSIN Liu et al. [2024a], as well as SAM-based adaptations Chen et al. [2025b], Hu et al. [2022], demonstrating its effectiveness in handling ambiguous and complex referring scenarios.

As illustrated in Fig. 3, EviSAM3 produces more accurate and complete segmentation masks across targets of varying scales under ambiguous referring expressions. In contrast, competing methods often exhibit imprecise boundaries or miss small regions, highlighting the advantage of our progressive evidence integration.

4.3 Ablation Study (RQ2)

Component-wise Ablation Analysis. Table 3 shows that vanilla SAM3 (without EIM and ED-MoE) performs poorly on both datasets, highlighting its limitation in handling ambiguous referring expressions. Adding EIM brings substantial gains across all metrics, demonstrating the effectiveness of progressive evidence integration for improving cross-modal alignment. Further introducing ED-MoE yields consistent improvements, especially at higher IoU thresholds, indicating enhanced refinement of fine-grained details under ambiguity. The larger gains on RefSegRS further suggest its advantage in handling semantically ambiguous expressions in the RRSIS task. Notably, the

Table 3: Component-wise ablation analysis on RRSIS-D and RefSegRS datasets.

		EIM	ED-MoE	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
RRSIS-D	val	-	-	11.95	10.69	8.91	6.72	3.97	15.56	12.82
		✓	-	79.01	72.99	61.70	48.51	27.64	77.63	67.70
		✓	✓	83.45	77.82	66.95	53.33	32.36	81.38	71.25
	test	-	-	11.17	9.82	8.24	6.43	3.88	14.66	11.64
		✓	-	79.04	73.60	62.04	47.22	27.44	77.59	67.57
		✓	✓	82.81	77.65	66.37	50.56	30.82	80.27	70.65
RefSegRS	val	-	-	30.16	20.19	9.28	2.09	1.16	26.02	24.71
		✓	-	91.18	89.10	84.69	76.33	24.83	79.51	79.47
		✓	✓	98.14	96.29	94.90	93.27	83.53	92.44	91.32
	test	-	-	26.53	18.00	7.59	2.42	0.77	28.03	25.69
		✓	-	80.02	73.97	64.45	46.23	10.68	75.46	68.65
		✓	✓	89.93	85.25	79.58	68.63	45.95	86.51	80.56

Table 4: EMA momentum coefficient analysis on the RefSegRS dataset.

μ	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
$\mu=0.0$	86.50	83.23	71.83	80.51	79.53
$\mu=0.3$	97.22	94.05	82.53	91.23	90.34
$\mu=0.6$ (Ours)	98.14	94.90	83.53	92.44	91.32
$\mu=0.9$	97.81	96.79	81.96	90.84	89.83

Table 5: Ablation study of evidence length and dimension on the RefSegRS dataset.

	E_t	E_d	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
Tokens	4	256	96.06	90.95	70.30	89.61	88.66
	8	256	98.14	94.90	83.53	92.44	91.32
	16	256	96.54	92.81	74.64	90.07	89.70
	32	256	95.79	91.88	71.23	89.89	88.95
Dims	8	128	95.82	91.42	72.62	90.16	88.47
	8	256	98.14	94.90	83.53	92.44	91.32
	8	512	97.91	94.90	80.28	91.84	90.43
	8	1024	96.06	93.04	75.02	90.89	89.87

improvements become more pronounced at stricter IoU thresholds, reflecting more accurate boundary delineation enabled by accumulated evidence.

Design-level Ablation Analysis. We further analyze key design choices in EviSAM3. As shown in Table 4, momentum-based updates are essential: $\mu=0.0$ performs poorly, while a moderate value ($\mu=0.6$) achieves the best overall performance by balancing stability and adaptability in evidence accumulation. A larger momentum ($\mu=0.9$) improves certain metrics but degrades others, indicating over-smoothing that weakens responsiveness to new evidence. Table 5 examines the capacity of the evidence representation. Performance improves up to $E_t=8$ tokens and $E_d=256$, but declines with larger settings, suggesting that excessive evidence introduces redundancy and noise, which hinders effective aggregation. These results indicate that a moderate evidence capacity is crucial for reliable reasoning.

4.4 Evidence Integration Analysis (RQ3)

Figure 4 illustrates the progressive evidence accumulation process in EviSAM3. While the base path initially produces coarse and spatially dispersed responses, the ED-MoE branch progressively refines predictions by integrating complementary multi-granularity evidence from both visual and linguistic cues. As the inference depth increases, the evidence responses become increasingly concentrated and better aligned with the semantic intent of the referring expression. This progressive refinement effectively suppresses irrelevant background regions while enhancing target-consistent activations. Ultimately, the aggregated evidence map exhibits a stable and compact focus on the object region, clearly demonstrating the effectiveness of the proposed evidence-driven reasoning mechanism for fine-grained referring remote sensing segmentation.

As shown in Fig. 5a, the heatmaps summarize 400 uniformly sampled RRSIS-D validation cases over a 5×5 grid of object area ratio and prompt length, where the left panel reports mean ΔIoU and the right shows mean IoU. The **consistently positive ΔIoU across all bins** demonstrates that the proposed **Evidence Integration** mechanism reliably improves the baseline, with gains most pronounced for small-to-medium objects ($a \in [0.1\%, 2\%]$) and longer prompts ($\ell \geq 6$), where progressive integration of multiple evidences is most beneficial. In contrast, large objects ($a > 10\%$) with short prompts ($\ell \leq 3$) show smaller Δ but already achieve high IoU ($\gtrsim 0.7$), indicating

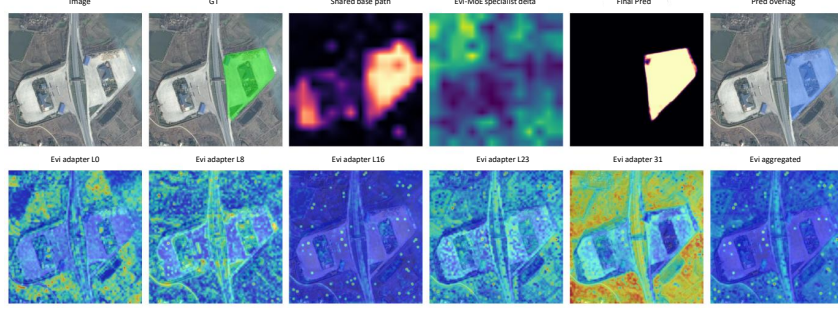
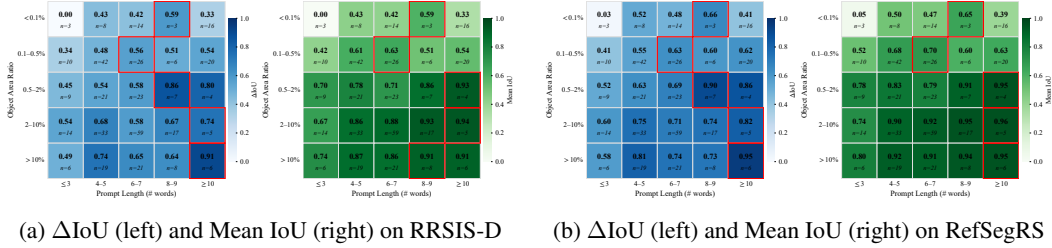


Figure 4: Visualization of evidence accumulation in EviSAM3 for the query “a service-area-like region toward the right”, showing progressively refined and more concentrated responses.



(a) Δ IoU (left) and Mean IoU (right) on RRSIS-D (b) Δ IoU (left) and Mean IoU (right) on RefSegRS

Figure 5: Conditional performance analysis across prompt length and object area ratio. Each cell reports IoU (or Δ IoU), with n indicating the number of samples in the bin.

a saturation effect, and **no negative Δ is observed**, suggesting consistent improvement without degrading the baseline. Similar trends are observed on RefSegRS in Fig. 5b.

4.5 Failure Case Analysis (RQ4)

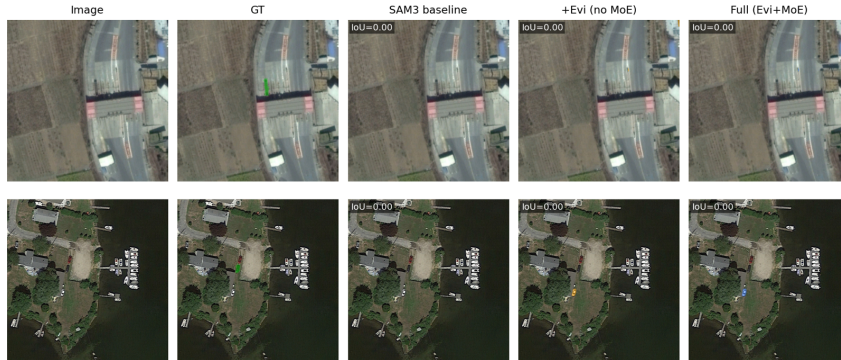


Figure 6: Failure cases of EviSAM3 on ambiguous queries. Two representative low-IoU examples illustrate backbone-limited failure and MoE mis-routing, corresponding to the queries “a gray vehicle” (top) and “the vehicle slightly larger than the one in the upper-left” (bottom).

As shown in Fig. 6, we present representative failure cases of EviSAM3 selected from a 200-sample validation pool with $\text{IoU}_{\text{Full}} \leq 0.30$. In the backbone-limited case, all variants fail to localize the target due to weak visual responses for small or occluded objects, resulting in the absence of a reliable base map. In contrast, the MoE mis-routing case shows that although +Evi partially captures the target, the Full model is distracted by a visually similar region, suggesting misaligned routing weights w . These failure modes are complementary, indicating limitations in both upstream visual representation and downstream routing, and point to potential directions for further improvement.

5 Conclusion

We present **EviSAM3**, an evidence-driven framework for referring remote sensing image segmentation that addresses the mismatch between concept-level prompting in SAM3 and the need for fine-grained referring grounding. By modeling grounding as a progressive evidence accumulation process, EviSAM3 incrementally integrates visual–linguistic cues through an evidence memory mechanism and refines predictions via evidence-aware expert prompting. This design enables the model to progressively suppress irrelevant regions while strengthening semantically consistent evidence across iterative reasoning stages.

Limitation and Future Work. EviSAM3 may fail when the backbone cannot provide reliable initial evidence, especially for small or occluded targets. In addition, the expert routing in ED-MoE can be suboptimal in highly confusing scenes. Future work will explore stronger visual encoders, more robust evidence routing strategies, and tighter coupling between evidence modeling and foundation models. Extending the framework to temporal settings, such as video-based referring segmentation with dynamic evidence accumulation, is another promising direction.

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